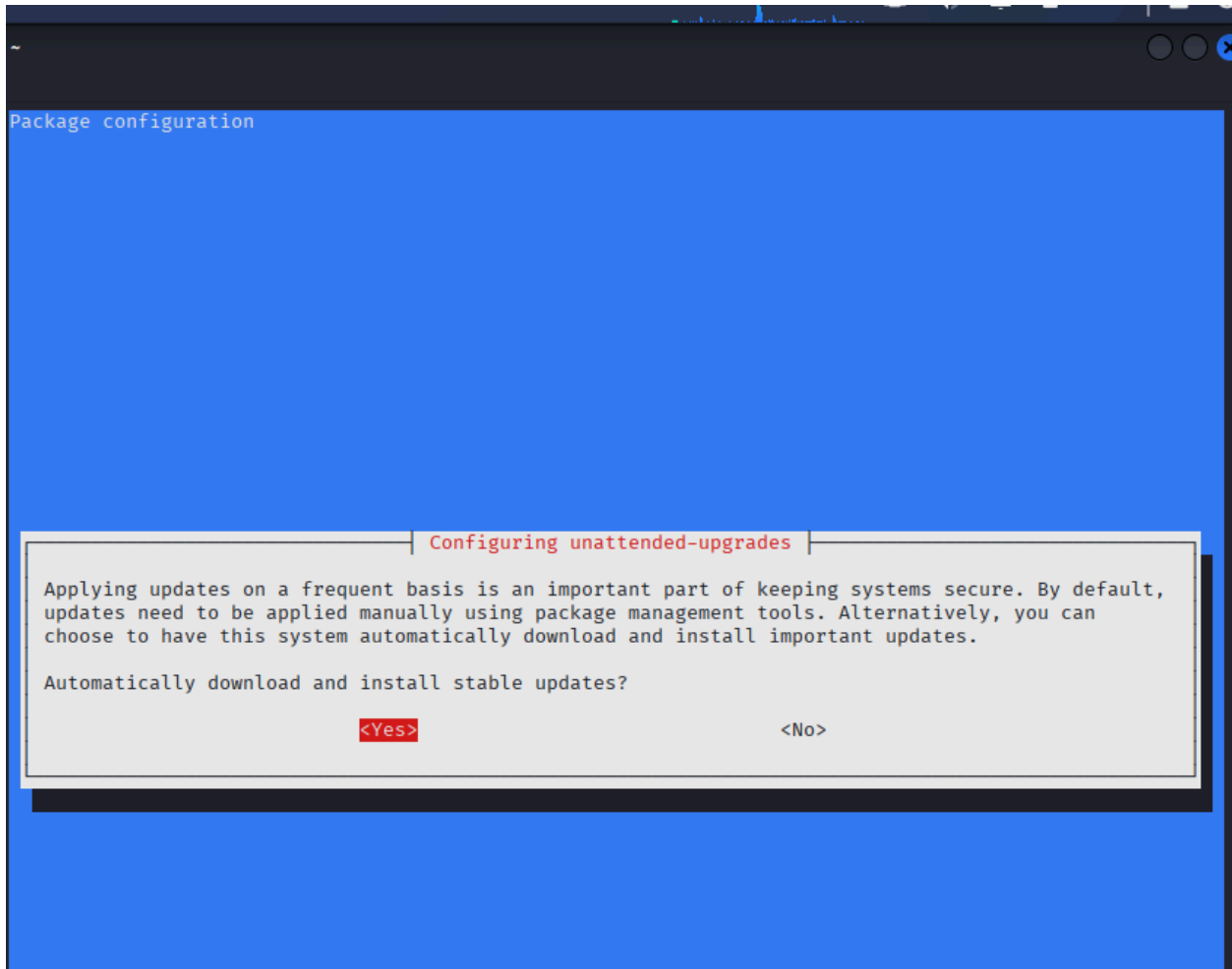




The system was successfully updated using `apt update` and `apt upgrade`. The `unattended-upgrades` package was installed and automatic stable updates were enabled to help keep the system secure.

Implementing Two-Factor Authentication (2FA) with Google Authenticator

Step 1: Installed Google Authenticator

1. Update the system package index:
`sudo apt update`
2. Install the Google Authenticator PAM module:
`sudo apt install libpam-google-authenticator -y`

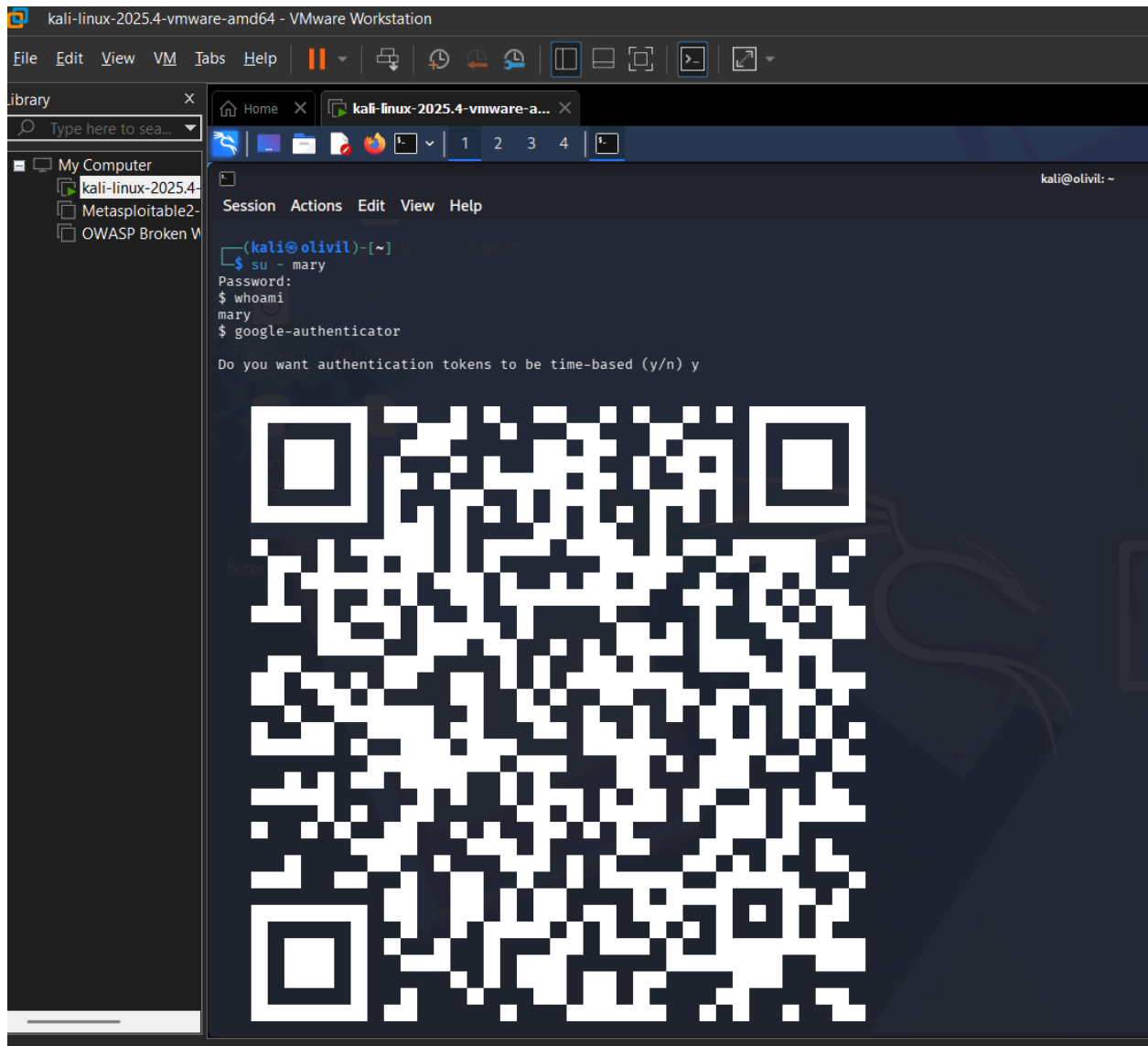
The screenshot shows a VMware Workstation window with a Kali Linux virtual machine. The terminal is open, and the user is running the following commands:

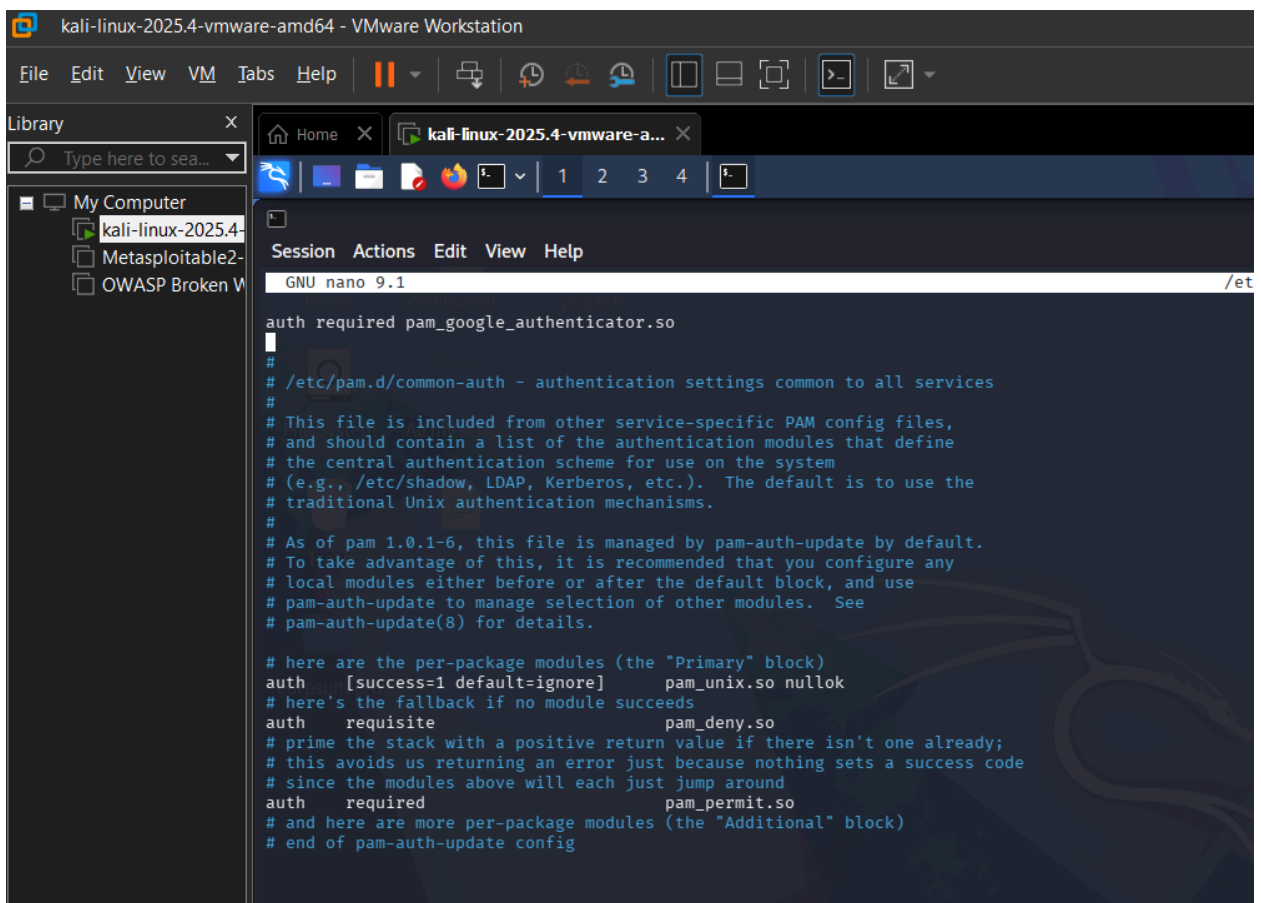
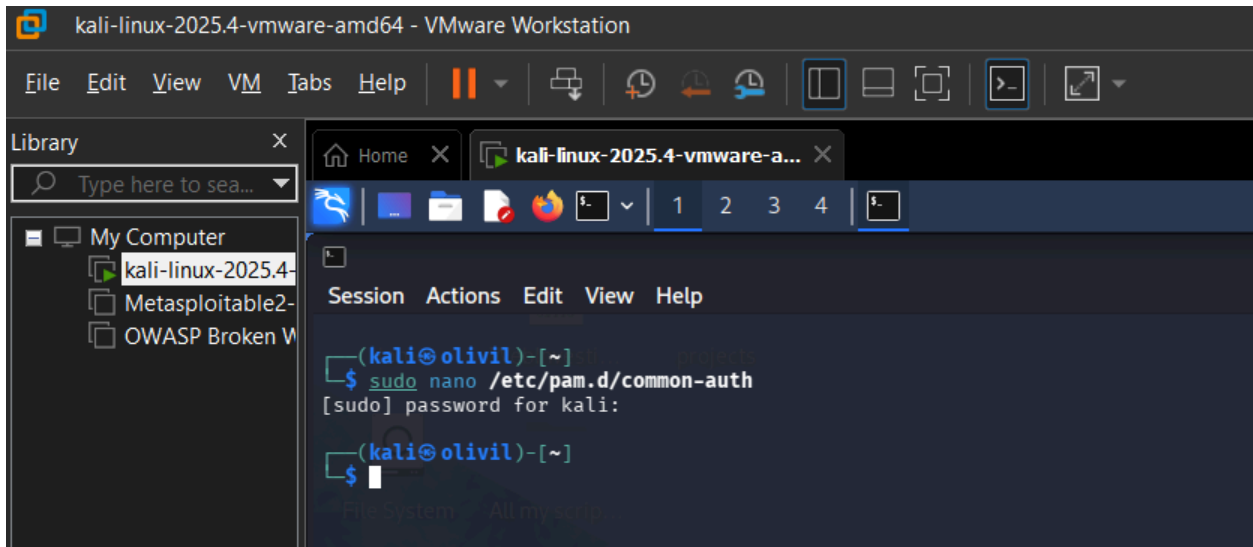
```
(kali@olivil)-[~]  
$ sudo apt update  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Ign:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb)  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [889 kB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Fetched 18.6 MB in 4min 56s (62.8 kB/s)  
24 packages can be upgraded. Run 'apt list --upgradable' to see them.  
  
(kali@olivil)-[~]  
$ sudo apt install libpam-google-authenticator -y  
Installing:  
  libpam-google-authenticator  
  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 24  
  Download size: 46.9 kB  
  Space needed: 132 kB / 51.5 GB available  
  
Get:1 http://kali.download/kali kali-rolling/main amd64 libpam-google-authenticator amd64 20250213-1.11-0.1 [46.9 kB]  
Fetched 46.9 kB in 24s (1,960 B/s)  
Selecting previously unselected package libpam-google-authenticator.  
(Reading database... 454924 files and directories currently installed.)  
Preparing to unpack ../libpam-google-authenticator_20250213-1.11-0.1_amd64.deb...  
Unpacking libpam-google-authenticator (20250213-1.11-0.1)...  
Setting up libpam-google-authenticator (20250213-1.11-0.1)...  
Processing triggers for Kali-menu (2026.3.2)...  
Processing triggers for man-db (2.13.1-1)...  
Scanning processes ...  
Scanning linux images ...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
  
(kali@olivil)-[~]  
$
```

2. Run the Google Authenticator setup:
google-authenticator

3. Follow the prompts:

- QR Code: Scan the QR code using the Google Authenticator app on your smartphone.
- Emergency Codes: Save the emergency backup codes provided (these can be used if you lose access to the app).
- Enable Time-Based Tokens: Answer yes when asked if you want to enable time-based tokens.

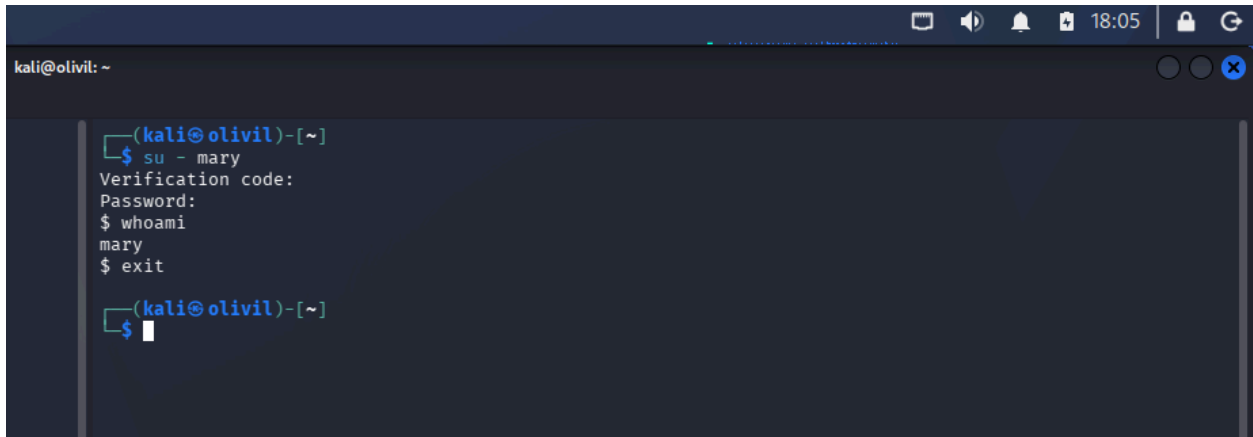




The PAM configuration was successfully updated to include Google Authenticator, enabling 2FA authentication for the system.

Step 4: Test 2FA Configuration

1. Open a new terminal or SSH session and attempt to log in as the configured user.
2. After entering the username and password, you should be prompted for the 2FA token generated by the Google Authenticator app.

A terminal window titled 'kali@olivil: ~' with standard Linux window controls. The prompt is '(kali@olivil)-[~]'. The user enters 'su - mary'. The prompt changes to 'Verification code:'. The user enters a code. The prompt changes to 'Password:'. The user enters 'mary'. The prompt changes to '\$ whoami'. The output is 'mary'. The user enters '\$ exit'. The prompt returns to '(kali@olivil)-[~]'.

```
(kali@olivil)-[~]
$ su - mary
Verification code:
Password:
$ whoami
mary
$ exit
(kali@olivil)-[~]
$
```

Mary successfully logged in by providing both her Google Authenticator verification code and password, confirming that the 2FA setup is working correctly

Implementing Key-Based Authentication and Securing SSH Access

Scenario

You are a cybersecurity analyst at SecureTech Ltd. In addition to disabling root login, the company wants to implement key-based authentication for enhanced SSH security. Your Exercise is to configure SSH for key-based authentication, test the setup, and ensure the root user is securely disabled.
